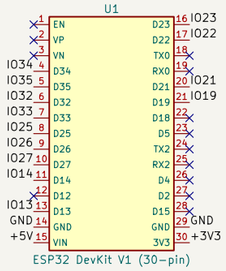
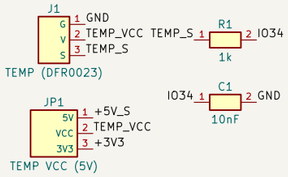
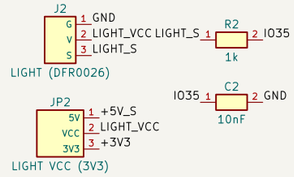
ESP32 DevKit V1 (30-pin) — sockets for the DevKit



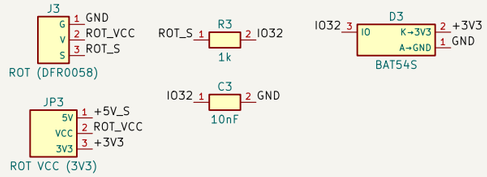
TEMP — DFR0023 — GPIO34 — VCC default: 5V



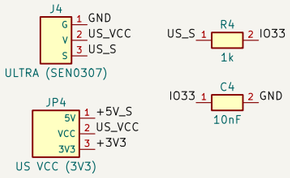
LIGHT — DFR0026 — GPIO35 — VCC default: 3V3



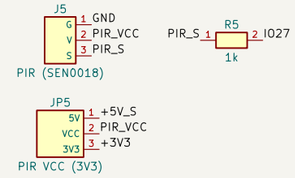
ROT — DFR0058 — GPIO32 — VCC default: 3V3



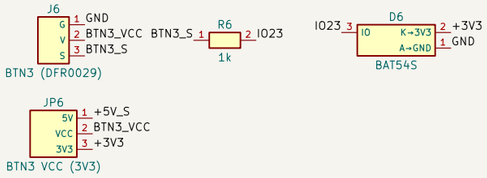
ULTRA — SEN0307 — GPIO33 — VCC default: 3V3



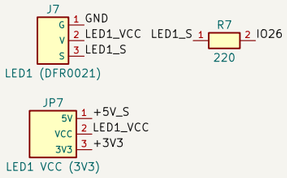
PIR — SEN0018 — GPIO27 — VCC default: 3V3



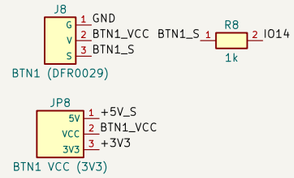
BTN3 — DFR0029 — GPIO23 — VCC default: 3V3



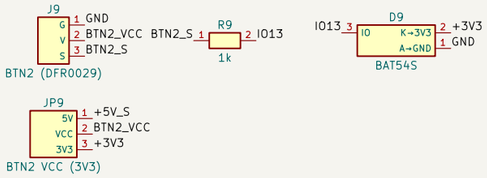
LED1 — DFR0021 — GPIO26 — VCC default: 3V3



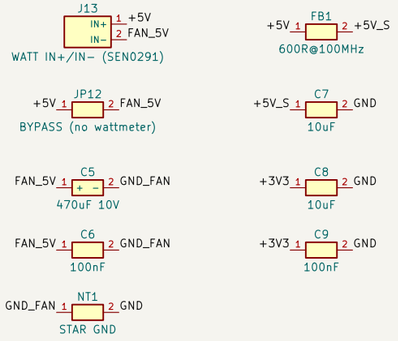
BTN1 — DFR0029 — GPIO14 — VCC default: 3V3



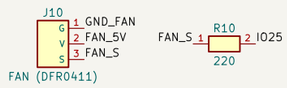
BTN2 — DFR0029 — GPIO13 — VCC default: 3V3



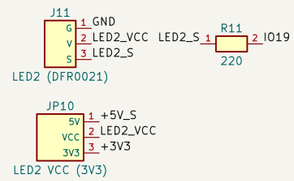
POWER — fan path is metered and has its own ground back to the star point



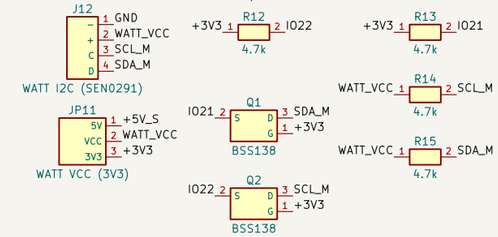
FAN — DFR0411 — GPIO25 — VCC default: metered 5V



LED2 — DFR0021 — GPIO19 — VCC default: 3V3



WATTMETER I2C — SEN0291 — SDA GPIO21 / SCL GPIO22 — VCC default: 3V3



BATTERY — regulated 5 V (power bank, 4x NiMH) — SS34 blocks reverse polarity and USB back-feed



Notes

1. Each module VCC jumper: cap on 5V+VCC = 5 V, cap on VCC+3V3 = 3.3 V. Defaults: TEMP on 5V, the rest on 3V3.
2. Every signal goes through a series resistor and a BAT54S clamp, so a 5 V module cannot lift a GPIO above ~3.6 V.
3. Wattmeter I2C goes through a BSS138 level shifter, so the SEN0291 may run on 5 V or 3V3.
4. The fan takes 5 V only through the wattmeter (IN+ -> IN-). Close BYPASS only if the wattmeter is not fitted.
5. Fan ground (GND_FAN) joins logic GND only at NT1, next to the DevKit GND pin (star ground).
6. Sensor 5 V (+5V_S) is filtered from VIN by FB1, so fan noise stays off the sensor rail.

Author: Mohamed Khamis

H1 M3 H2 M3 H3 M3 H4 M3

Pins: docs/esp32-wiring.md, firmware/main/board_pins.h
ESP32 DevKit V1 carrier for the DFRobot Gravity modules
Mohamed Khamis
Sheet: /
File: smart-room-carrier.kicad_sch

Title: Smart Room carrier board

Size: A3	Date: 2026-10-20	**Rev: 1.1**
KiCad E.D.A. 10.0.6		Id: 1/1